

# Distinct Values Subsequences

Time limit: 1.00 s

Memory limit: 512 MB

Given an array of  $n$  integers, count the number of subsequences where each element is distinct.

A subsequence is a sequence of array elements from left to right that may have gaps.

## Input

The first line has an integer  $n$ : the array size.

The second line has  $n$  integers  $x_1, x_2, \dots, x_n$ : the array contents.

## Output

Print the number of subsequences with distinct elements. The answer can be large, so print it modulo  $10^9 + 7$ .

## Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq x_i \leq 10^9$

## Example

| Input        | Output |
|--------------|--------|
| 4<br>1 2 1 3 | 11     |

*Explanation:* The subsequences are  $[1]$  (two times),  $[2]$ ,  $[3]$ ,  $[1, 2]$ ,  $[1, 3]$  (two times),  $[2, 1]$ ,  $[2, 3]$ ,  $[1, 2, 3]$  and  $[2, 1, 3]$ .